The empirical recurrence for OEIS A223325, proved by transfer matrix

Adrian Perez Fontelles
Independent researcher

4 September 2026

Abstract

OEIS A223325 counts the ways of writing the vertices of a fixed graph G into an array of width 5 so that any two cells adjacent in one of the named directions carry vertices joined by an edge of G; here G is icosahedron graph. The entry carries an empirical recurrence of order 32 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. A slice of the array would be a state, but there are 248832 legal slices here and that is too many to carry one by one. The condition names only the edge relation, so the automorphism group of G — of order 120 — acts on the arrays and commutes with the transfer matrix; the walk may then be run on the ORBITS of slices instead of on the slices, and $S = 2208$ of those suffice.

2020 Mathematics Subject Classification. 05C15, 05A15, 05C25, 15A18.

1 The conjecture

OEIS A223325 is "Rolling icosahedron footprints: number of 5Xn 0..11 arrays starting with 0 where 0..11 label vertices of an icosahedron and every array movement to a horizontal or antidiagonal neighbor moves along an icosahedral edge.". It has offset 1 and begins

$$20736, 1953125, 446265625, 101966340125, 24143758634125, 5759605530667625, 1379464144963464625, 33081750$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 410*a(n-1) -51839*a(n-2) +2998354*a(n-3) -88929070*a(n-4) +1198807214*a(n-5) +2889121905*a(n-6) -340756682572*a(n-7) +4538793109678*a(n-8) -13887817317962*a(n-9) -242674670034177*a(n-10) +2612400275589524*a(n-11) -4250937817051838*a(n-12) -72579353969012690*a(n-13) +404852006352195731*a(n-14) +294243010726671144*a(n-15) -8038059311698199807*a(n-16) +14527240098893219908*a(n-17) +63207106281330811172*a(n-18) -241285400707856322796*a(n-19) -99441316590244134158*a(n-20) +1499939986936247507922*a(n-21) -1264768034332853154292*a(n-22) -3709575277042038429058*a(n-23) +6249708500546392632287*a(n-24) +1712083980241985550010*a(n-25) -7806673210840206689815*a(n-26) +1552725322135658220450*a(n-27) +3587994096679078794377*a(n-28) -1272290169371896871744*a(n-29) -473135759143977735225*a(n-30) +224855787210470741790*a(n-31) -20543329382473731600*a(n-32) for n>36

As of the "Last modified" line on the live entry (Oct 03 2025 21:31:11, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

Let G be the graph on the label set $\{0, 1, \dots, 11\}$ whose vertices are the twelve vertices of a regular icosahedron, two of them adjacent when an edge joins them. That is the icosahedron graph: 12 vertices, 5-regular, 30 edges.

An array x is a map from the cells of a rectangle to the vertices of G. The entry names the directions antidiagonal and horizontal, that is the cell offsets

$$(1,-1),(0,1),$$

and asks that whenever two cells of the array differ by one of those offsets, the two vertices they carry are joined by an edge of G.

Write $\Gamma = \mathrm{Aut}(G)$ for the group of adjacency-preserving bijections of the vertex set. It was computed by backtracking over the vertices in breadth-first order, assigning each an image consistent with every vertex already placed, and has order $|\Gamma| = 120$.

Lemma 1. *For $\sigma \in \Gamma$ the array $\sigma \circ x$ is valid whenever x is, and this is an action of Γ on the set of valid arrays. In particular the count does not depend on how the vertices of G are labelled.*

Proof. The condition mentions the vertices only through the relation "joined by an edge", which σ preserves in both directions. The same argument with an isomorphism between two labelled copies of G gives the second statement. $\square$

3 The cell the entry fixes

The entry asks for $x(0,0) = 0$. That condition is removed here, and paid for once at the end.

Lemma 2. *Γ is transitive on the vertices of G. Consequently the valid arrays with $x(0,0) = v$ are equinumerous for every vertex v, and the number with $x(0,0) = 0$ is the number of valid arrays with no condition on any cell, divided by* 12.

Proof. Transitivity was checked directly on the group computed above: the orbit of one vertex is the whole vertex set. Given vertices v, w pick $\sigma \in \Gamma$ with $\sigma v = w$; then $x \mapsto \sigma \circ x$ is a bijection from the valid arrays with $x(0,0) = v$ onto the valid arrays with $x(0,0) = w$, with inverse given by σ^{-1}. The 12 fibres of $x \mapsto x(0,0)$ therefore all have the same size and partition the unrestricted count. $\square$

4 Orbits of slices

Take the array one *column* at a time, in the direction in which it grows. The entry writes the array with n as the number of COLUMNS, so the slice added at each step is a column; transposing exchanges the horizontal and vertical directions and carries each diagonal direction to itself. Every offset the entry names has its two cells either in the same *column* or in two consecutive ones, so a *column* is a state: let M be the 0/1 matrix indexed by the legal *column*s with $M_{uv} = 1$ when v may follow u, and let $\tau = \mathbf{1}$. The number of valid arrays of m *column*s is $\mathbf{1}^\top M^{m-1}\tau$.

There are 248832 legal *column*s, which is more than can be carried one at a time. They need not be.

Lemma 3. *Γ acts on the columns coordinatewise; $M_{uv} = M_{\sigma u\, \sigma v}$ for every $\sigma \in \Gamma$, and τ is fixed by Γ. Consequently $M^m\tau$ is constant on each Γ-orbit of columns, for every $m \geq 0$.*

Proof. That v may follow u is again a conjunction of statements "these two vertices are joined by an edge", so σ preserves it; and τ is the all-ones vector. Induct on m: if $w = M^m\tau$ satisfies $w_{\sigma u} = w_u$ then $(Mw)_{\sigma u} = \sum_v M_{\sigma u\, v} w_v = \sum_v M_{u\, \sigma^{-1} v} w_v = \sum_{v'} M_{uv'} w_{\sigma v'} = (Mw)_u$. $\square$

Let $\mathcal{O}$ be the set of orbits, r_O a representative of O, and

$$\widehat{M}_{O,O'} = \#\{\, v \in O' : M_{r_O v} = 1 \,\}.$$

By Lemma 3 the value $(M^m\tau)_{r_O}$ is what $M^m\tau$ takes on all of O, and one step reads

$$(Mw)_{r_O} = \sum_{O'} \widehat{M}_{O,O'}\, w_{r_{O'}},$$

so the whole walk may be run on $\mathcal{O}$. Summing over *columns* and grouping,

$$\mathbf{1}^\top M^m \tau = \sum_{O\in\mathcal{O}} |O|\,(M^m\tau)_{r_O} = \iota^\top \widehat{M}^m \mathbf{1}, \qquad \iota_O = |O|,$$

and the count the entry publishes is that divided by 12, by Lemma 2. Here $S = |\mathcal{O}| = 2208$, down from 248832.

The orbit representatives are generated directly rather than by listing the *columns* and grouping them. A *column* is built one cell at a time, carrying the set of σ that have so far written exactly the same prefix; a choice is abandoned as soon as one of them would write a smaller letter. What reaches the last cell is the lexicographically least member of its orbit, one per orbit, and the set still tied there is its stabiliser, so $|O| = |\Gamma|/|\Gamma_{r_O}|$ falls out of the same walk.

The walk index and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

5 The proof

Lemma 4. *Let $q(t) = t^{32} - \sum_i c_i t^{32-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^\top \widehat{M}^j q(\widehat{M})\mathbf{1}$ for the corresponding walk index j, so the recurrence holds from some point on if and only if that vanishes for all large j, and it is enough to examine $j < S$.*

Proof. Factor $\widehat{M}^j$ out of $\widehat{M}^{j+32} - \sum_i c_i \widehat{M}^{j+32-i}$. For the bound, $\widehat{M}^S$ is an integer combination of $I, \widehat{M}, \ldots, \widehat{M}^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^\top \widehat{M}^j q(\widehat{M})\mathbf{1}$ obey the monic recurrence given by its characteristic polynomial; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 5.

$$a(n) \;=\; 410\,a(n{-}1) - 51839\,a(n{-}2) + 2998354\,a(n{-}3) - 88929070\,a(n{-}4) + 1198807214\,a(n{-}5) + 2889121905\,a(n{-}$$

for every $n > 36$.

Proof. The vector $w = q(\widehat{M})\mathbf{1}$ was formed by 32 matrix–vector products in exact integer arithmetic and $\iota^\top \widehat{M}^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 36 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

6 Verification

Five checks, each independent of the algebra above.

First, the model reproduces all 11 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition, with the graph rebuilt by a different route—the cube from its corner points and face normals in space, the icosahedron as a pentagonal antiprism capped at both poles—and the arrays written out cell by cell, each cell tested against the cells it must be joined to. That recount applies the entry's own condition on the top-left cell directly, so it tests Lemma 2 as well: no orbit, no lumping and no division is involved in it, and the published terms came back.

Third, the automorphism group was checked to be a group—closed under composition and inverses—and every one of its 120 elements verified to preserve the edge set.

Fourth, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fifth, the annihilation test of Lemma 4 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A223325.

[2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[3] J. G. Kemeny and J. L. Snell, *Finite Markov Chains*, Springer, 1976. (Lumping, Section 6.3.)

[4] H. S. M. Coxeter, *Regular Polytopes*, 3rd ed., Dover, 1973.

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.